Computer Organization and Architecture
(Code: CS256)

Lecture-27: Internal Memory Design II

Course Faculty

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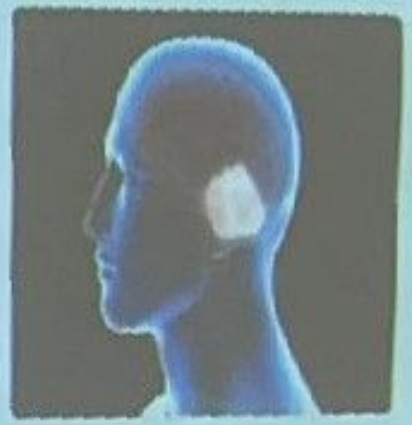
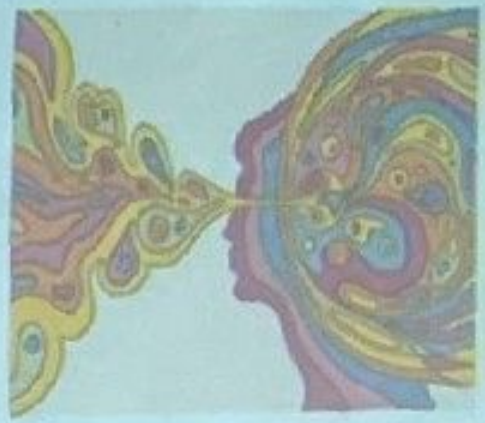
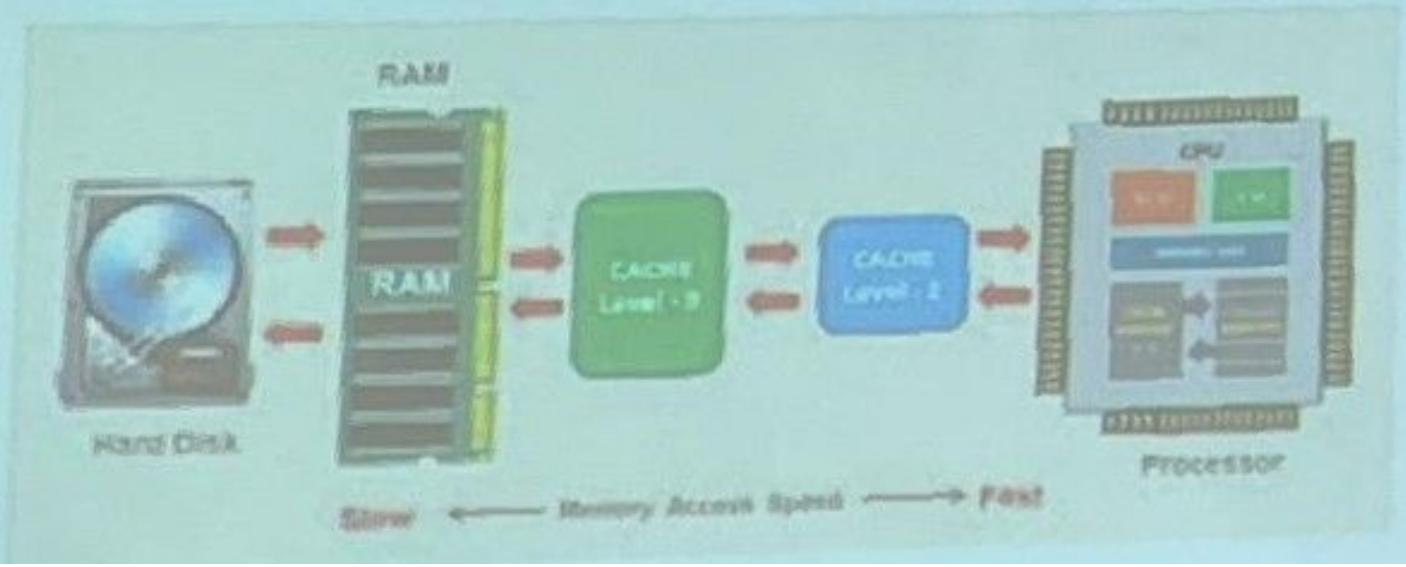
Location

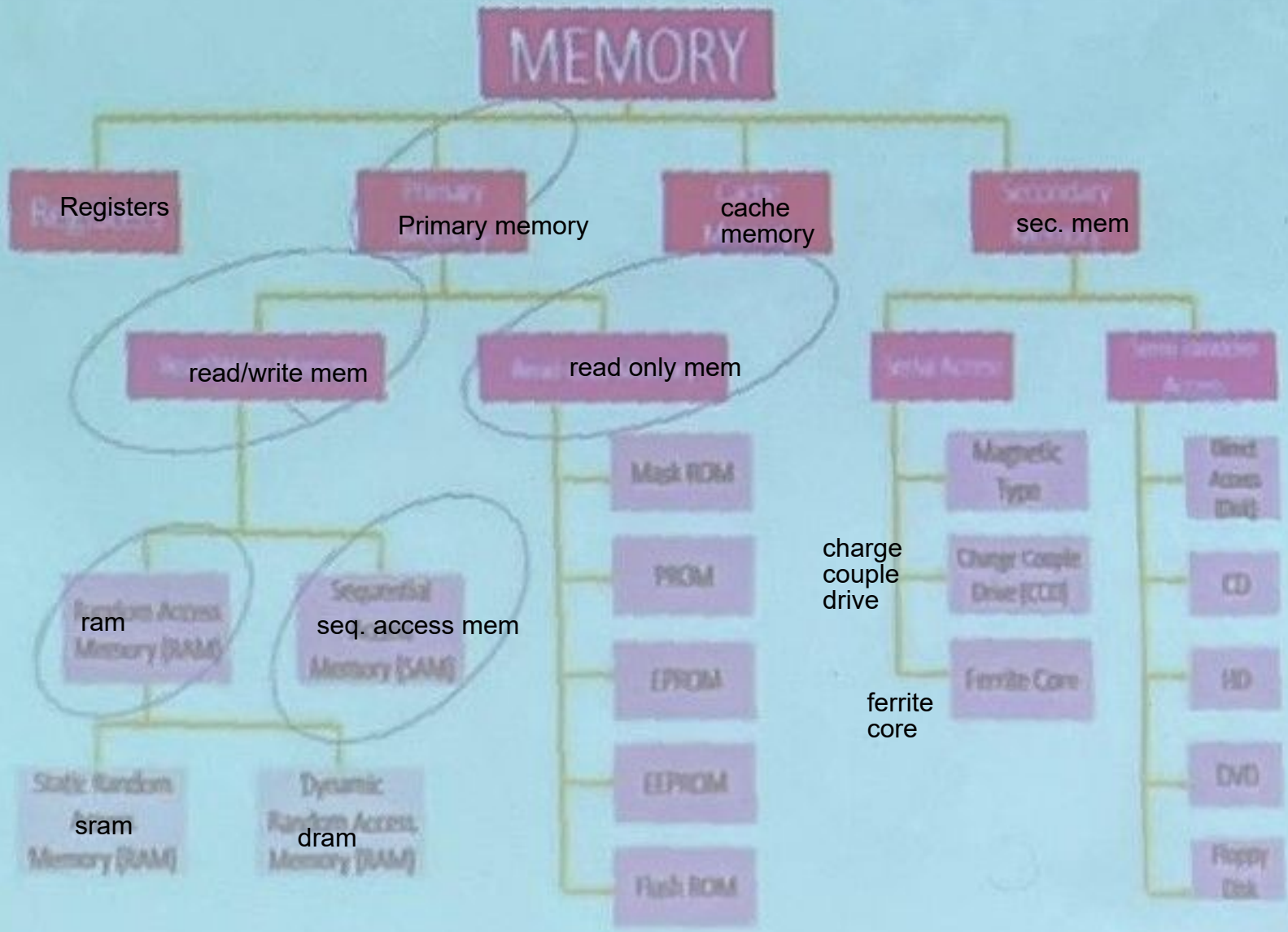
- CPU
  - Registers, Control Memory
- Internal
  - Cache
  - Main Memory – RAM
  - ROM
- External
  - Disk, Tape, …







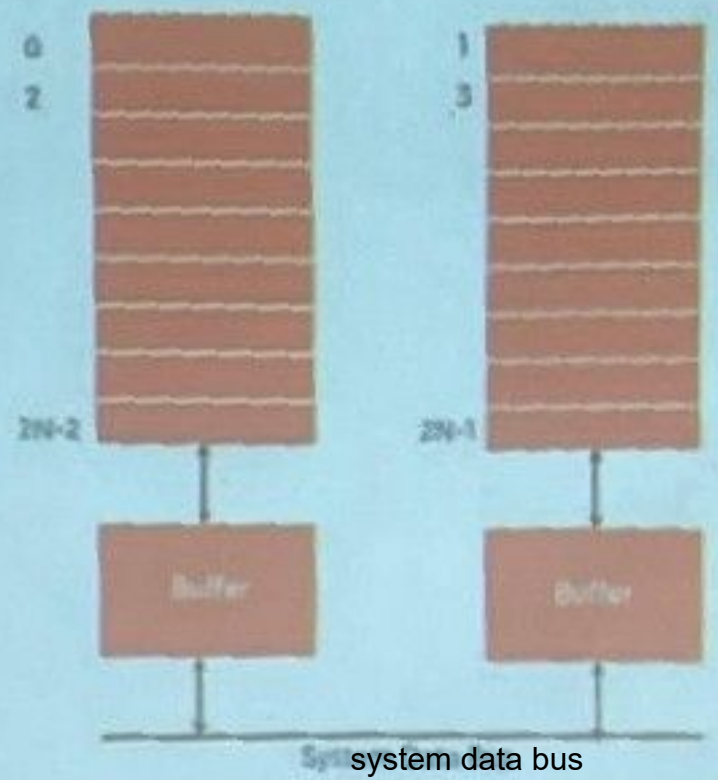
Memory Classification



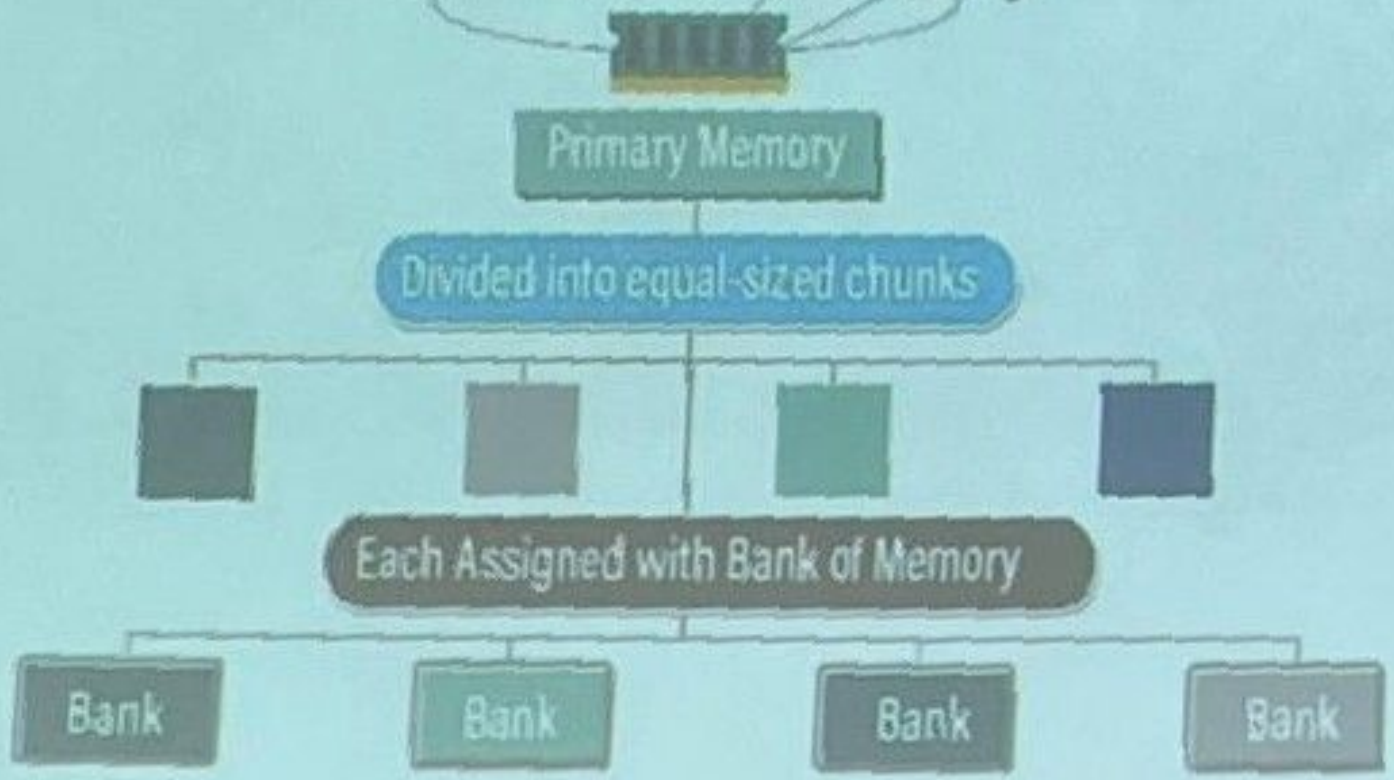
Interleaved Memory

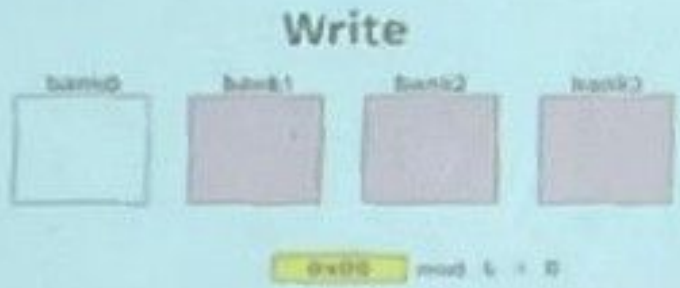
Interleaved Memory

- Main memory is collection of DRAM chips.
- A number of chips can be grouped together to form a *memory bank*.
- Banks independently service read or write requests.
- K banks can service K requests simultaneously increasing memory read or write rates by a factor of K.
- If consecutive words of memory are stored in different banks, then the transfer of a block of memory is speeded up.
- *It is possible to organize the memory banks in a way known as interleaved memory.*
- Each bank is now independently able to service.
- Banks are interleaved to enhance performance by reducing the latency between read and write operations.
- *It provides better memory bandwidth, allowing the processor to access data from different memory banks simultaneously.*



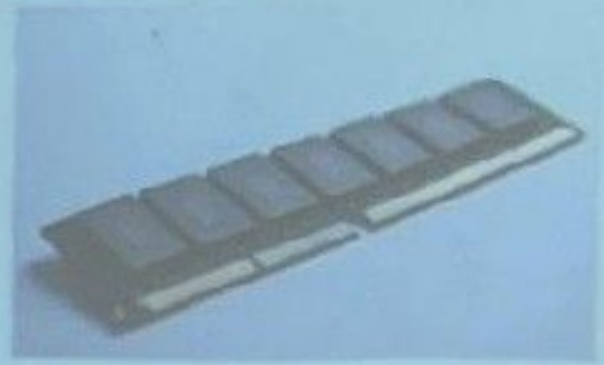
Interleaved Memory





Why Interleaved Memory?

- ***Overcoming Single Module Limitations:***
  - Traditional memory systems rely on a single memory module to store and retrieve data, which can create a bottleneck due to limited data access speed.
  - It divides memory into multiple banks, allowing simultaneous access to increase memory bandwidth and enhance data transfer rates.
- ***Reducing Memory Latency:***
  - In non-interleaved memory systems, the CPU often waits for a memory module to complete its ongoing operation before accessing data.
  - It mitigates this latency issue by enabling the CPU to access data from different memory modules simultaneously, reducing waiting times and improving overall system performance.



Types of Interleaved Memory

- High-Order Interleaving Memory
- Low-Order Interleaving Memory

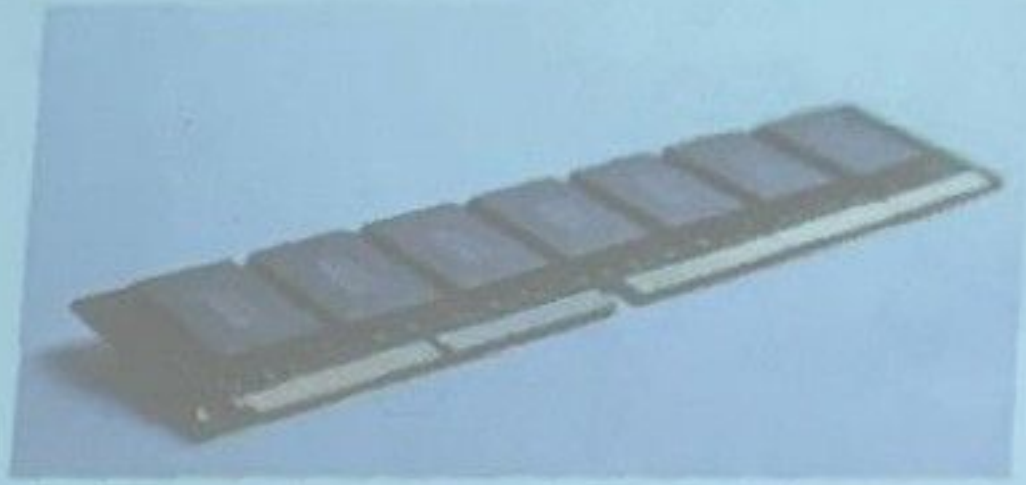
Module 0	Module 1	Module 2	Module 3	Module 4	Module 5	Module 6	Module 7
0	4	8	12	16	20	24	28
1	5	9	13	17	21	25	29
2	6	10	14	18	22	26	30
3	7	11	15	19	23	27	31

High-Order Interleaving

- High-order interleaving involves dividing memory into blocks and distributing data across these blocks.
- Each block contains a range of memory addresses, and data is systematically spread across these blocks.
- *High-order interleaving is common in systems with limited memory modules and relatively large data word sizes.*
- It ensures even distribution of data across memory banks, which reduces contention and enhances access speed.

Low-Order Interleaving

- Low-order interleaving distributes data word by word across memory modules.
- Each memory module receives one part of each data word, and the system accesses them sequentially to assemble the complete data word.
- *Low-order interleaving is suitable for systems with more memory modules, ensuring that data access is distributed efficiently across modules.*
- It is often used in high-performance computing environments where parallel access to smaller data units is critical for system speed.



High and Low Order Interleaving

- *In high order memory interleaving, the most significant bits of the memory address decides memory bank* where a particular location resides. But, *in low order interleaving the least significant bits of the memory address decides the memory banks.*

Address bits	25-22	21-0
Use	Bank Select	Address to the chip

Module 0	Module 1	Module 2	Module 3	Module 4	Module 5	Module 6	Module 7
0	4	8	12	16	20	24	28
1	5	9	13	17	21	25	29
2	6	10	14	18	22	26	30
3	7	11	15	19	23	27	31

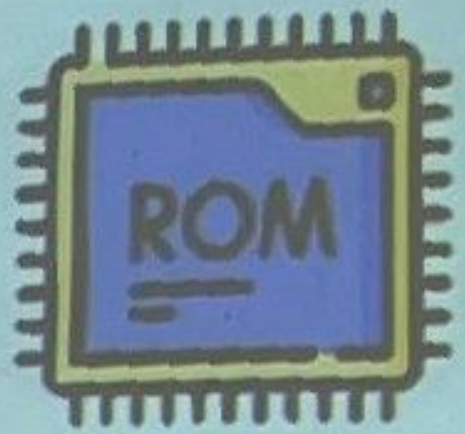
Address bits	25-4	3-0
Use	Address to the chip	Bank Select

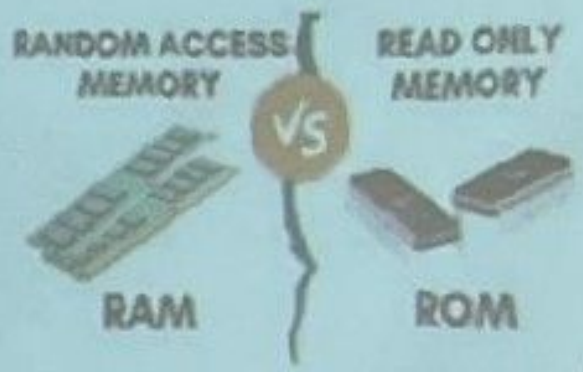
Module 0	Module 1	Module 2	Module 3	Module 4	Module 5	Module 6	Module
0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31

ROM

Read Only Memory

- *A read-only memory (ROM) is nonvolatile memory that contains a permanent pattern of data that cannot be changed.*
- While it is possible to read a ROM, it is not possible to write new data into it.
- A ROM like a RAM is a form of semiconductor memory technology.
- Many memory technologies lose the data once the power is removed.
- A ROM is nonvolatile. So, no power source is required to maintain the bit values in memory.
- As a result, this type of semiconductor memory technology is widely used for storing programs and data that must survive when a computer or processor is powered down.
- For example – if your pc is turned off the data in the ROM will not be erased.

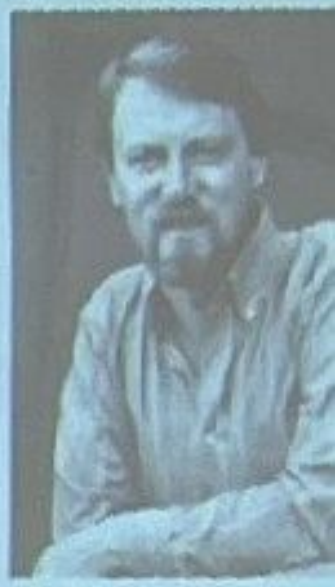




Use of ROM

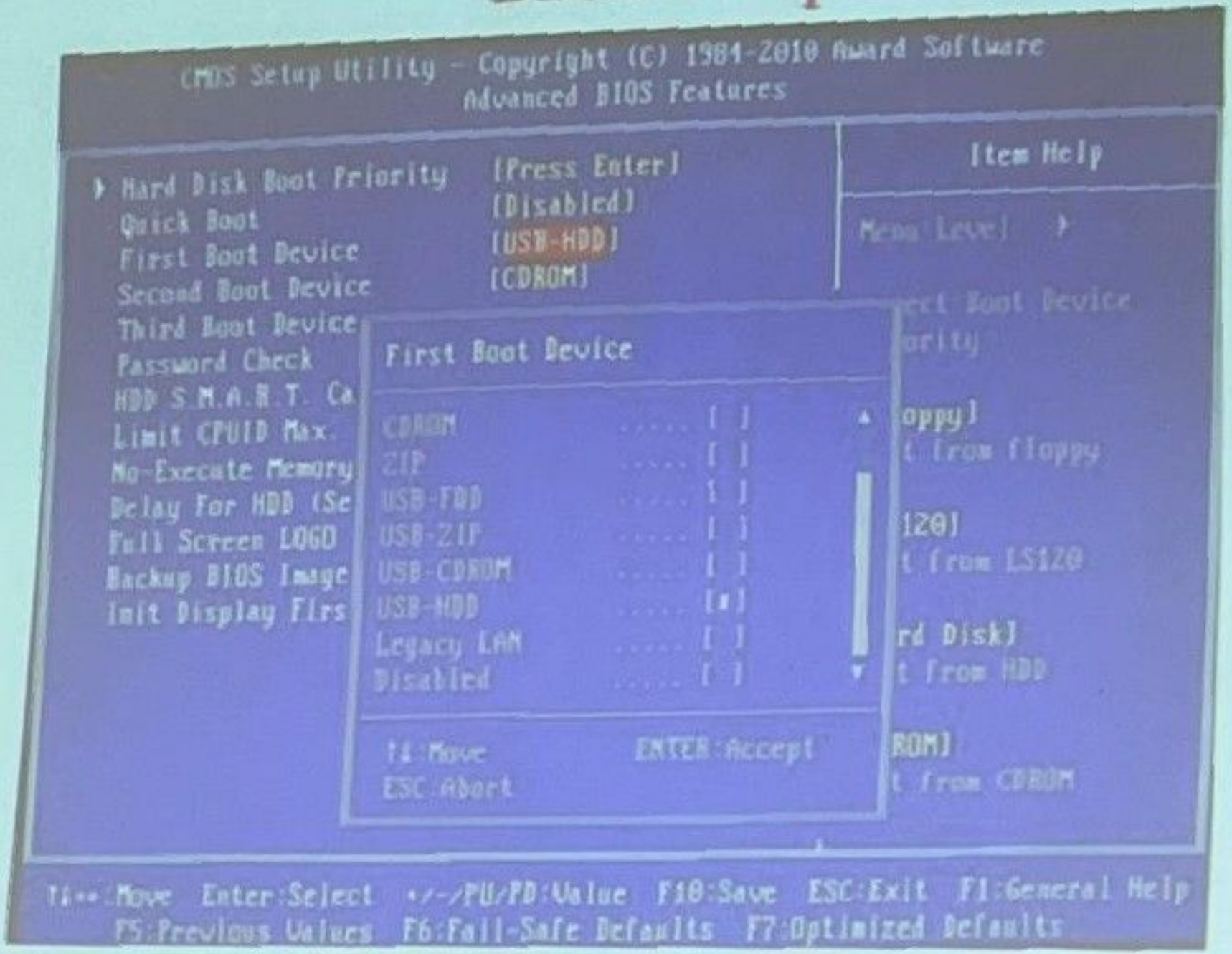
- *An important application of ROMs is microprogramming.*
- For example, ROM is used to store the BIOS (Basic input-output system) or firmware uses to start the computer system after it is powered on.
- *Therefore, all the important instructions that used to start the computer are programmed.*
- It also manages data flow between the computer's operating system (OS) and attached devices, such as the hard disk, video adapter, keyboard, mouse and printer.
- *Other potential applications include*
  - Library subroutines for frequently wanted functions
  - System programs
  - Function tables

BIOS



- *BIOS is a firmware embedded on the chip (ROM) on the computer's motherboard.*
- *It is also known as ROM BIOS, PC BIOS.*
- The term BIOS was first coined in 1975 by American computer scientist Gary Kildall.
- It was incorporated into IBM's first personal computer in 1981 and, in the years to come, gained popularity within other PCs, becoming an integral part of computers for some time.
- However, BIOS' popularity has waned in favor of a newer technology: *Unified Extensible Firmware Interface (UEFI).*

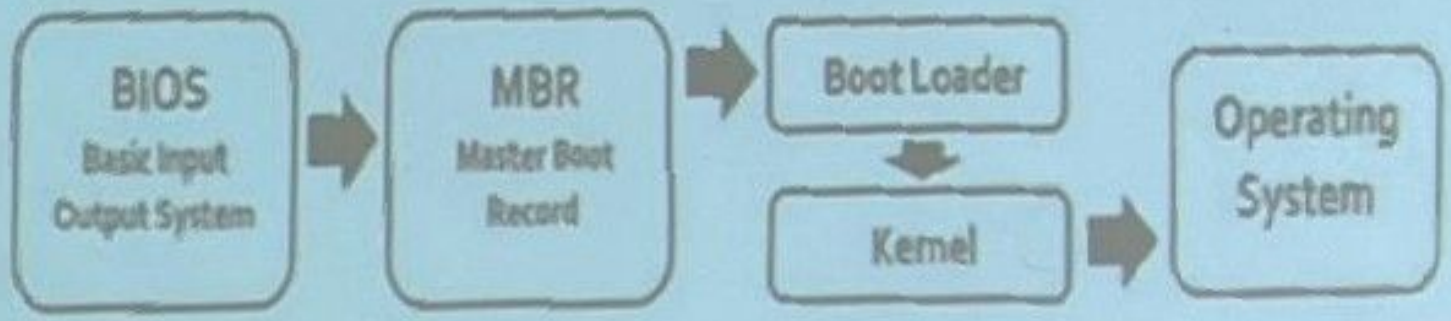
BIOS Setup


CMOS Setup Utility - Copyright (C) 1984-2010 Award Software
Advanced BIOS Features
Hard Disk Boot Priority [Press Enter]
Quick Boot [Disabled]
First Boot Device [USB-HDD]
Second Boot Device [CDROM]
Third Boot Device
Password Check
HDD S.M.A.R.T. Ca
Limit CPUID Max.
No-Execute Memory
Delay For HDD (Se
Full Screen LOGO
Backup BIOS Image
Init Display Firs
Item Help
Menu Level
First Boot Device
CDROM
ZIP
USB-FDD
USB-ZIP
USB-CDROM
USB-HDD
Legacy LAN
Disabled
↑↓:Move ENTER:Accept
ESC:Abort
↑↓→←:Move Enter:Select +/-/PU/PD:Value F10:Save ESC:Exit F1:General Help
F5:Previous Values F6:Fail-Safe Defaults F7:Optimized Defaults

Legacy BIOS

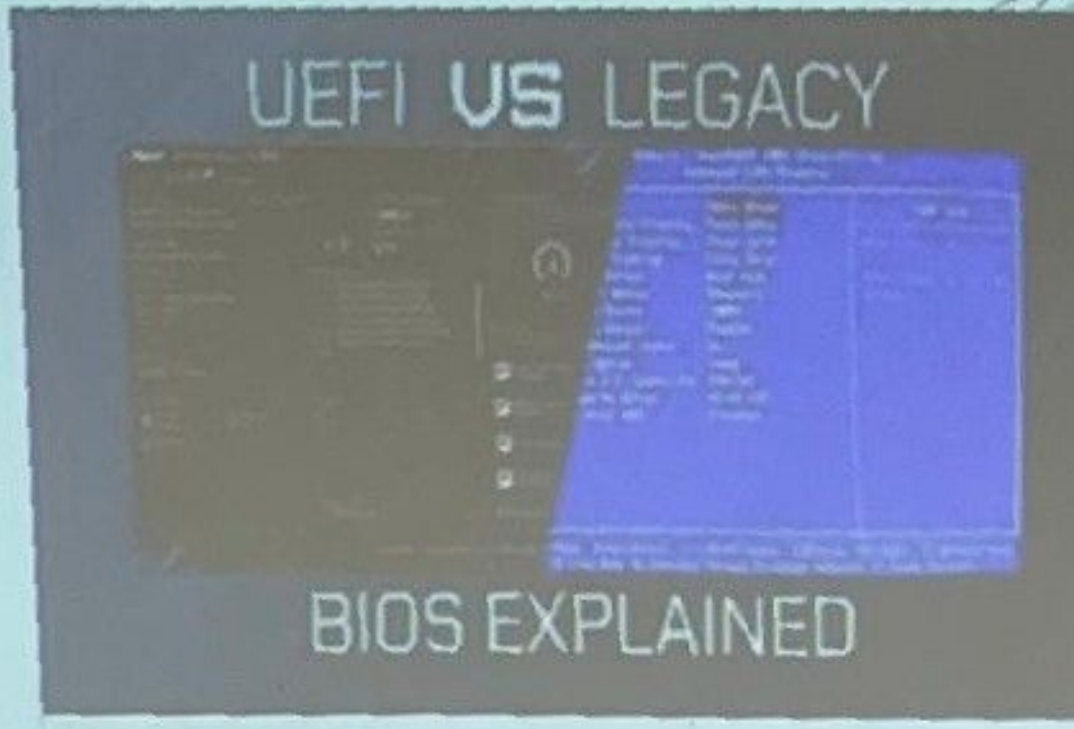
- *Legacy PC BIOS refers to the older, text-based firmware interface used to initialize and boot a computer.*
- *A BIOS is just considered a "Legacy" because it's a much older solution than a modern UEFI, but both will essentially function the same.*
- Some older hardware and software may not be compatible with UEFI,
- In fact, many older and less expensive motherboards still use the BIOS mode.
- Intel announced a plan in 2017 to retire support for legacy BIOS systems by 2020, replacing them with UEFI.

Legacy Boot



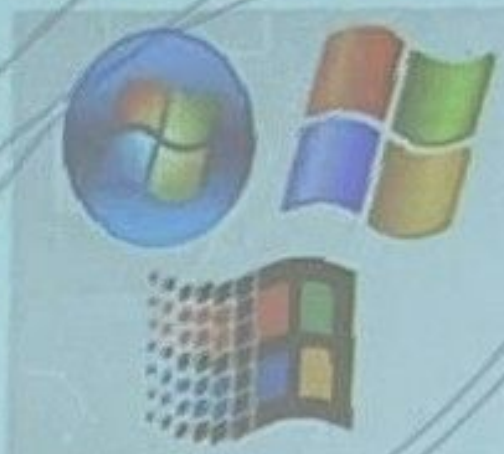
UEFI

- Nowadays, many users use UEFI boot to start up Windows as it has many significant advantages, like faster booting process and support for hard drives larger than 2 TB, more security features and so on.
- Although UEFI is more advanced and powerful than BIOS, It was not widespread and not all motherboards support the new firmware till 2020.
- UEFI possesses many new features and advantages that cannot be achieved through the traditional BIOS.
- It is aimed to completely replace the BIOS in the future.



Legacy BIOS vs UEFI

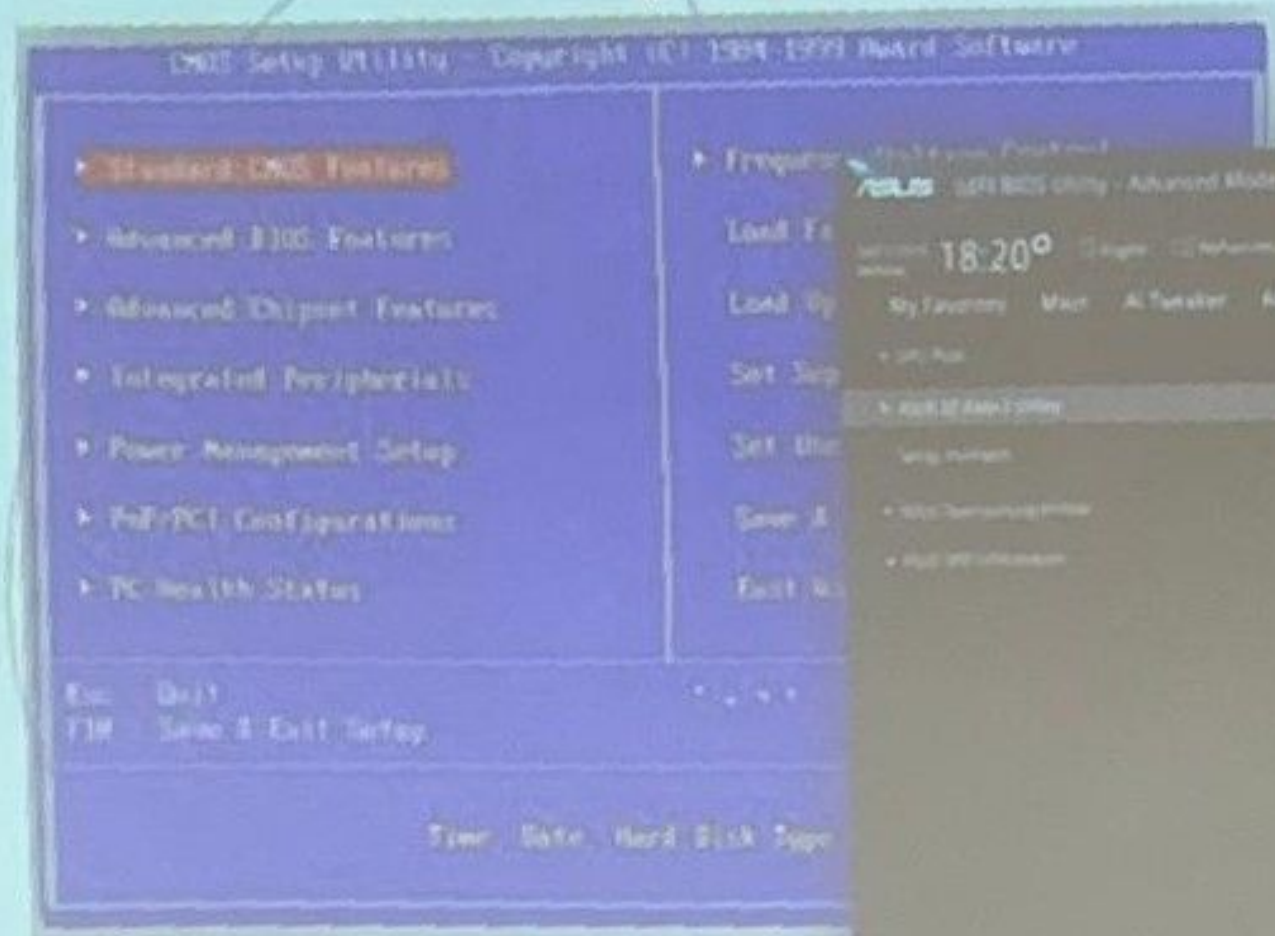
- Windows XP and its decade-plus of market dominance was entirely on the back of the humble BIOS, and had no compatibility to speak of with a fancy UEFI.



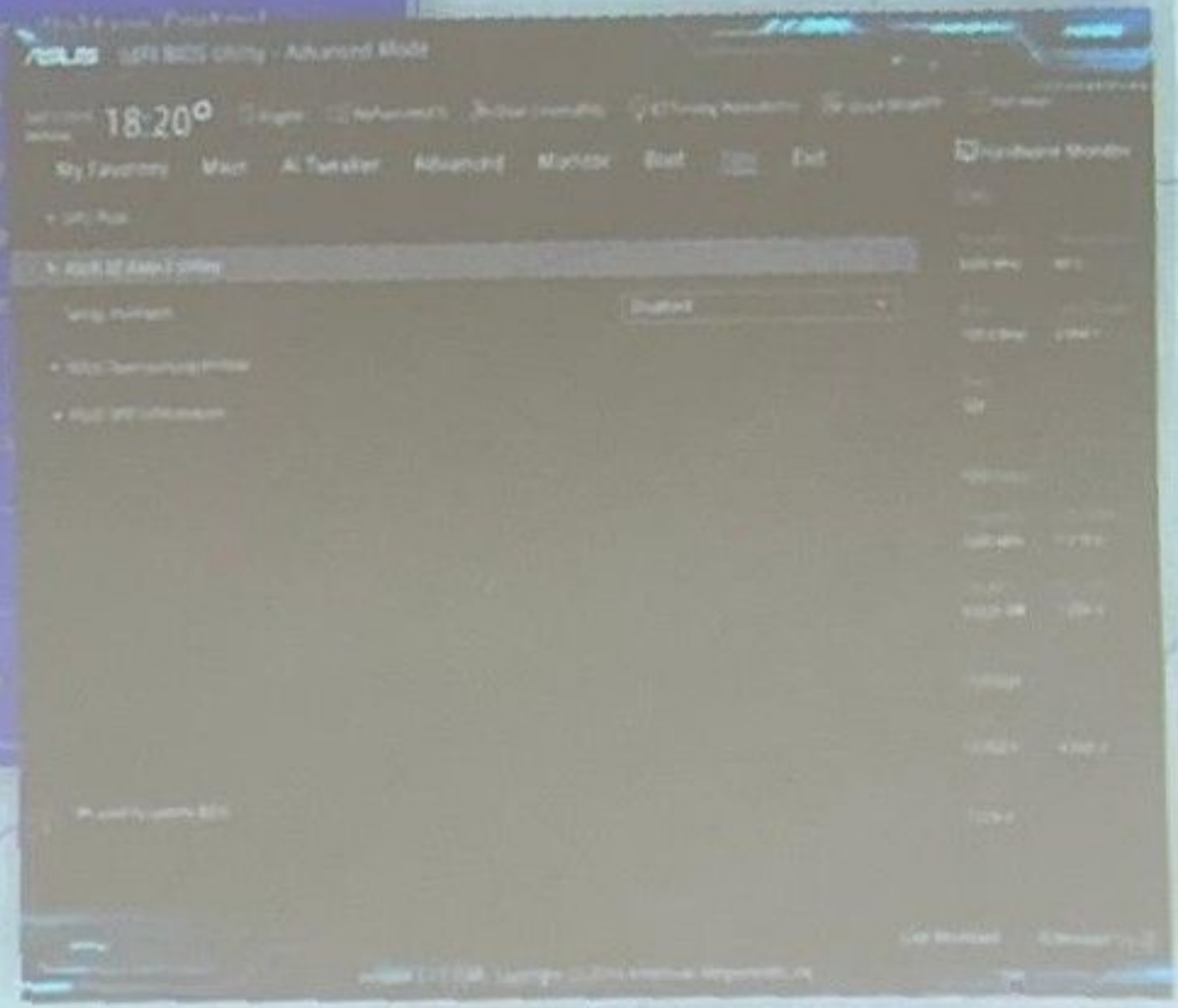
Windows Vista SP1
and Earlier Windows

32-Bit OS on 64-Bit
Firmware, or vice versa

Legacy and UEFI Setup

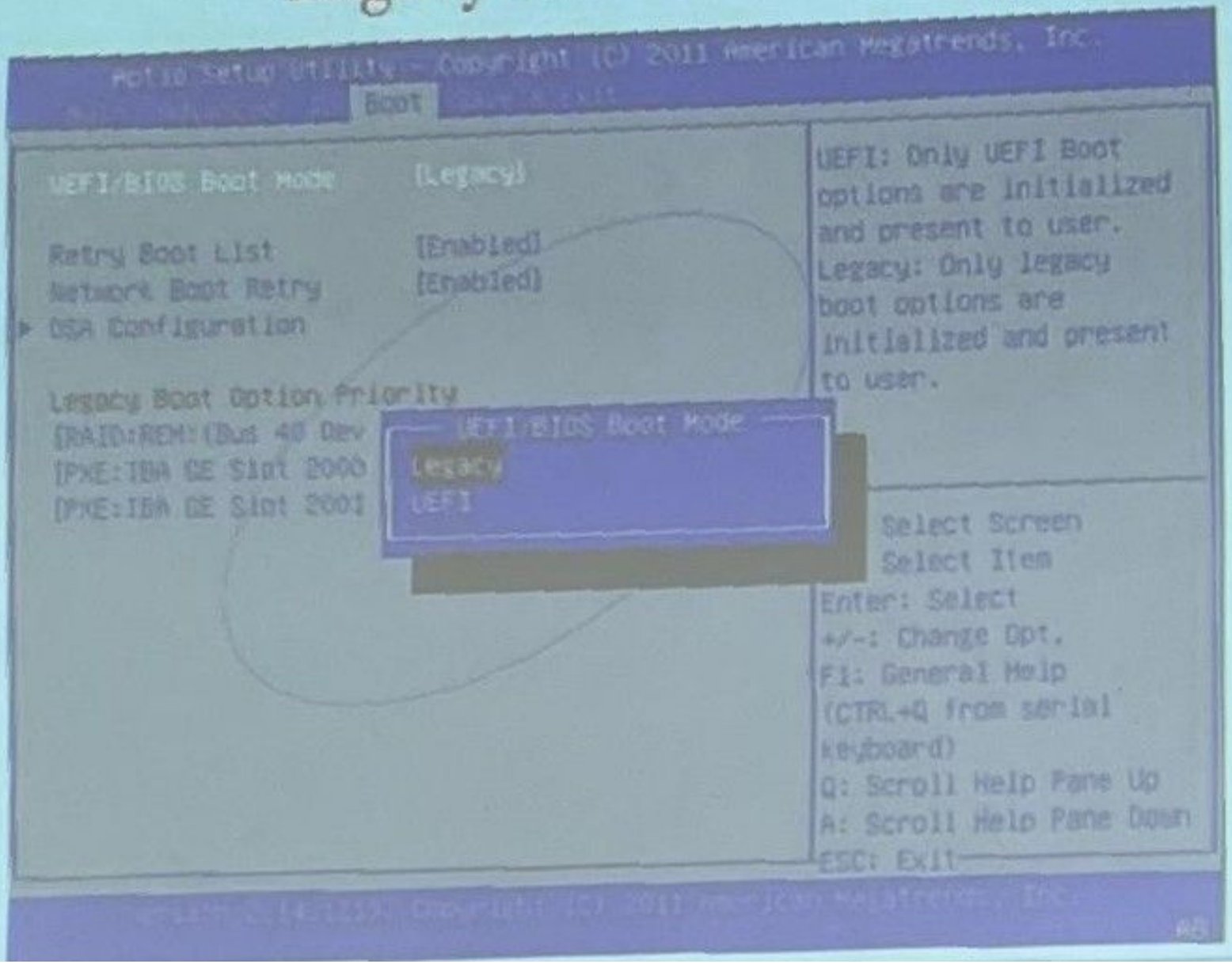


Legacy BIOS



UEFI BIOS

Legacy and UEFI Setup



Why UEFI ?

- *Is UEFI Always Better?*
- *If your hardware is compatible with UEFI, yes.*
- If you have the option to choose between UEFI and BIOS, you'll be best served by using UEFI.
- So, what improvements does UEFI have to offer over a Legacy BIOS implementation? Let's break it down:
  - Support for storage drives larger than 2 Terabytes.
  - Improved graphical interface and mouse support compared to Legacy BIOS.
  - Improved hardware support and compatibility compared to Legacy BIOS.
  - And finally, generally faster and more secure, all-around, than a Legacy BIOS.

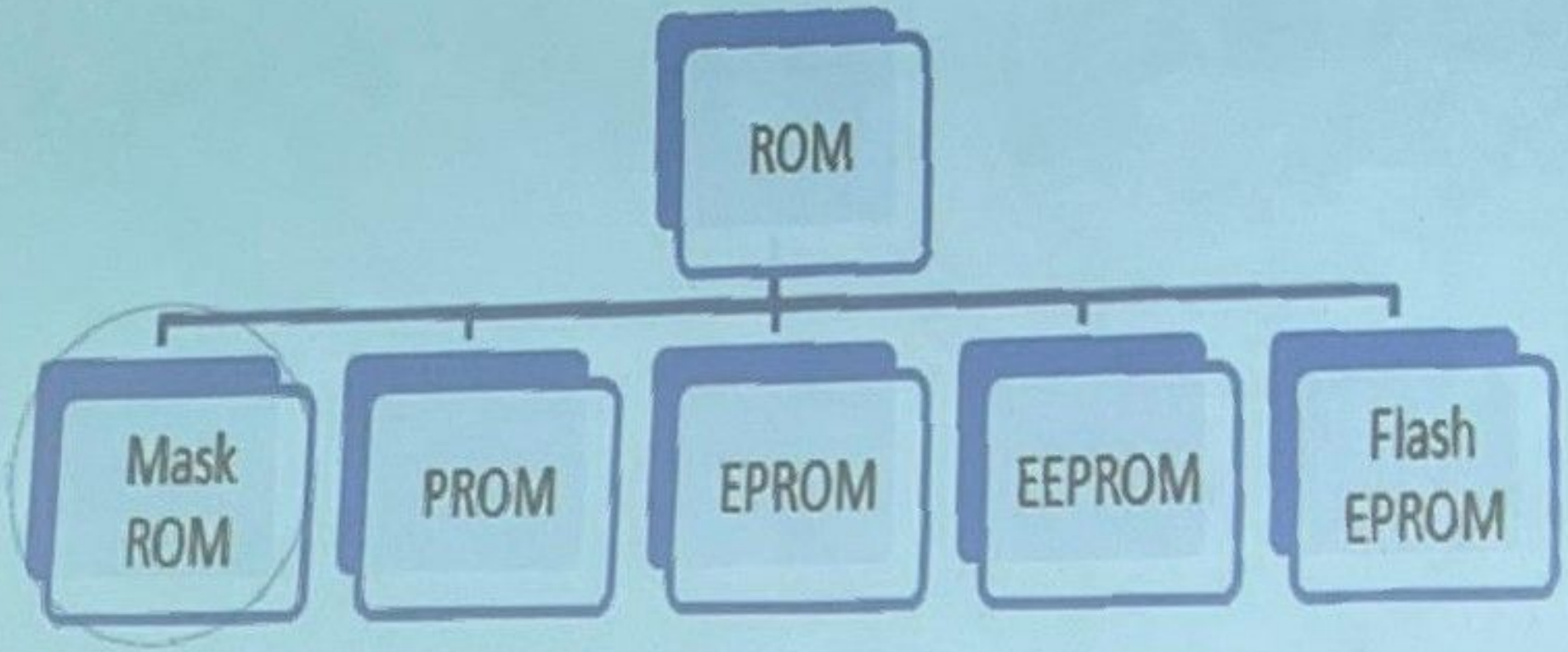
ROM Advantages

- The advantage of ROM is that the data or program is permanently in primary memory.
- Its static nature means it does not require refreshing.
- It needs never be loaded from a secondary storage device.
- It is easy to test.
- Different Memory testing algorithms are used.
- ROM Memory consumes low power.
- ROM memory is not expensive than RAM.

ROM Disadvantages

- The data insertion step includes a relatively large fixed cost, whether one or thousan copies of a particular ROM are fabricated.
- Inability to store large amounts of information.
- Unlike a RAM, ROM is a slower type of Memory.
- Once the data is stored in the ROM, we cannot modify, delete or overwrite the dat ROM.
- *There is no room for error.*
- If one bit is wrong, the whole batch of ROMs must be thrown out.

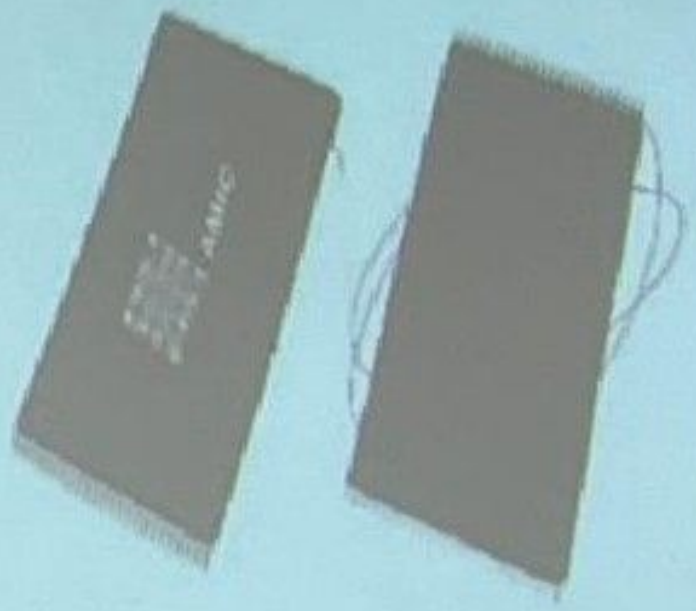
ROM Classification

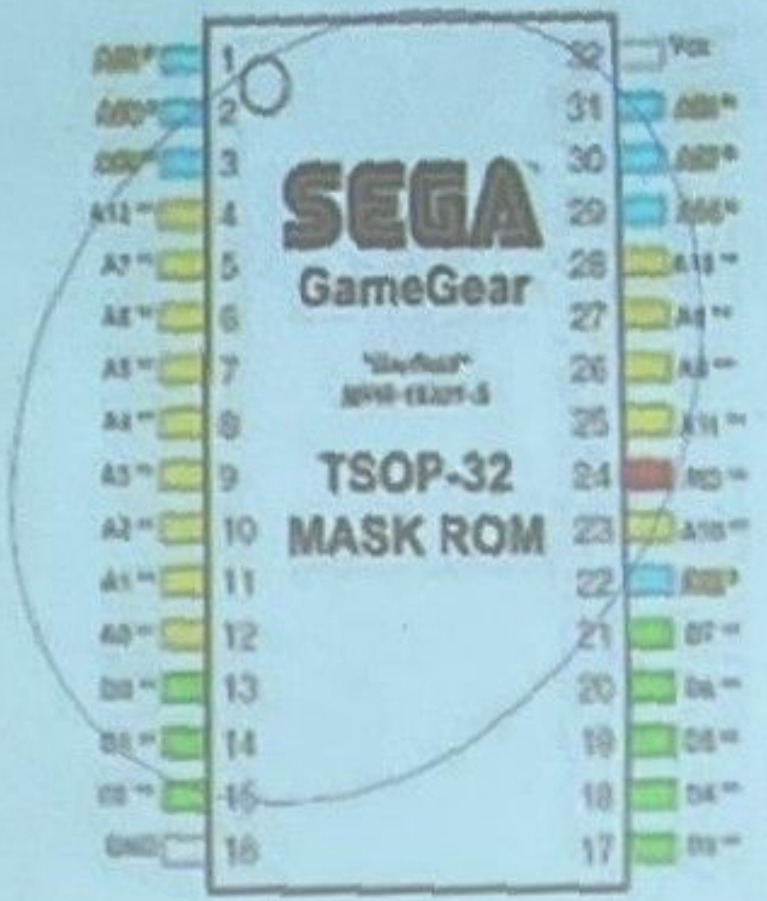


Mask ROM

- Mask ROM (MROM) is a type of ROM, that is masked off during production.
- "*Mask*" refers to parts of an integrated circuit to produce Mask ROM (MROM) chips during the design phase of the semiconductor fabrication process.
- The circuit is a thin electronic circuit for processing data and is covered with opaque plates called photomasks.
- Mask ROM (MROM) chips contain a software mask that is burned onto the chip during the design phase of the semiconductor manufacturing process.
- Examples of computer systems that use Mask ROM chips or MROM chips include network operating systems and server operating systems.
- In high volumes, mask read-only memory (ROM) is a cost-effective alternative to other ROMs.
- Like other types of ROM, mask ROM cannot enable the user to change the data stored in it.
- If it can, the process would be difficult or slow.

Mask ROM

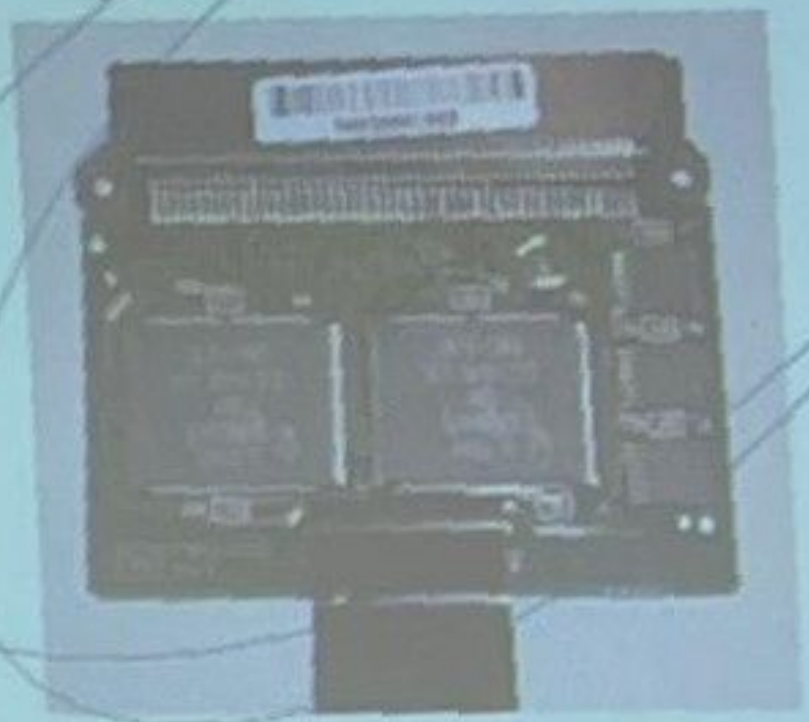


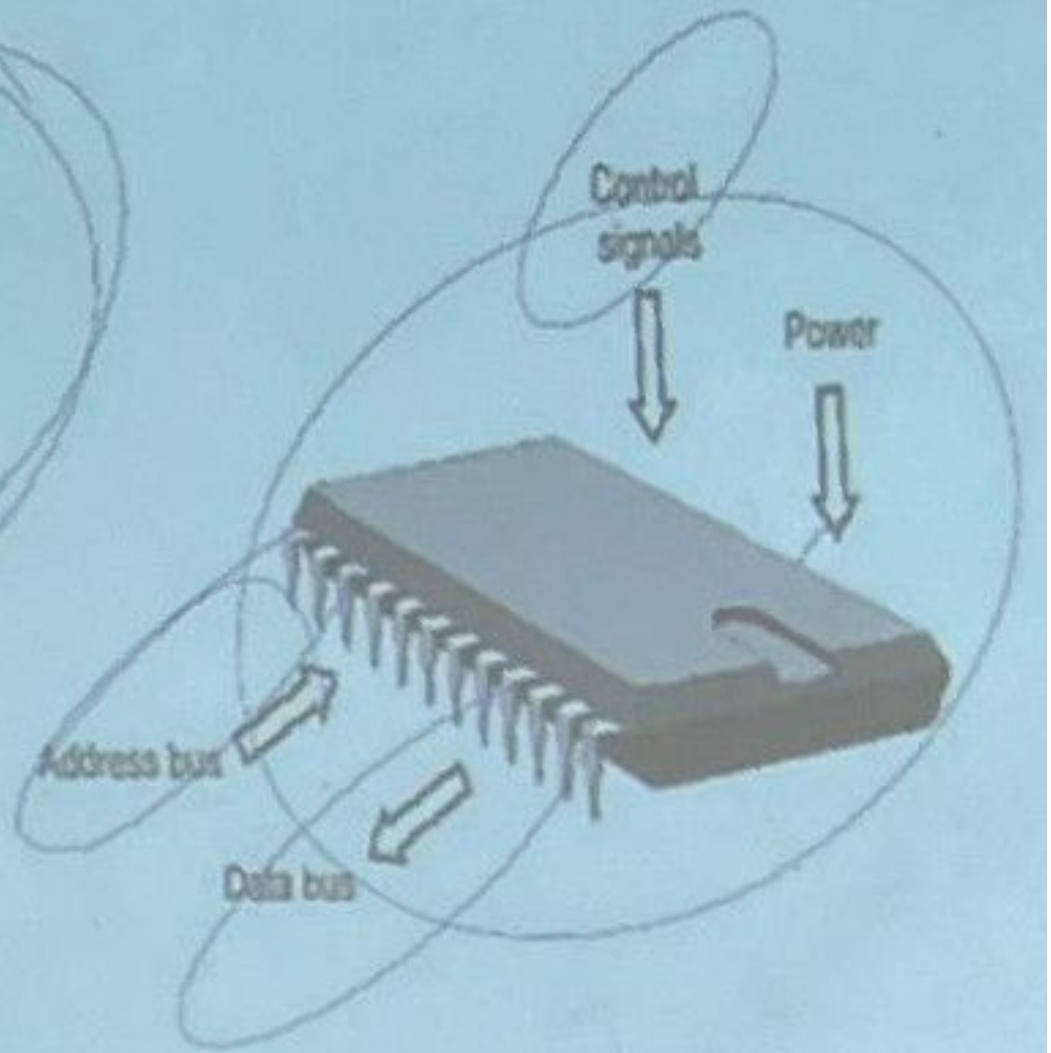


PROM

- *A Programmable Read Only Memory (PROM) is a semiconductor memory where the contents can be changed once after manufacture of the device.*
- The PROM was invented in 1956 by Wen Tsing Chow.
- PROMs may be used where the volume required does not make a factory-programmed ROM economical.
- *The PROM stores its data as a charge on a capacitor.*
- There is a charge storage capacitor for each cell and this can be read repeatedly as required.
- However it is found that after many years the charge may leak away and the data may be lost.

PROM





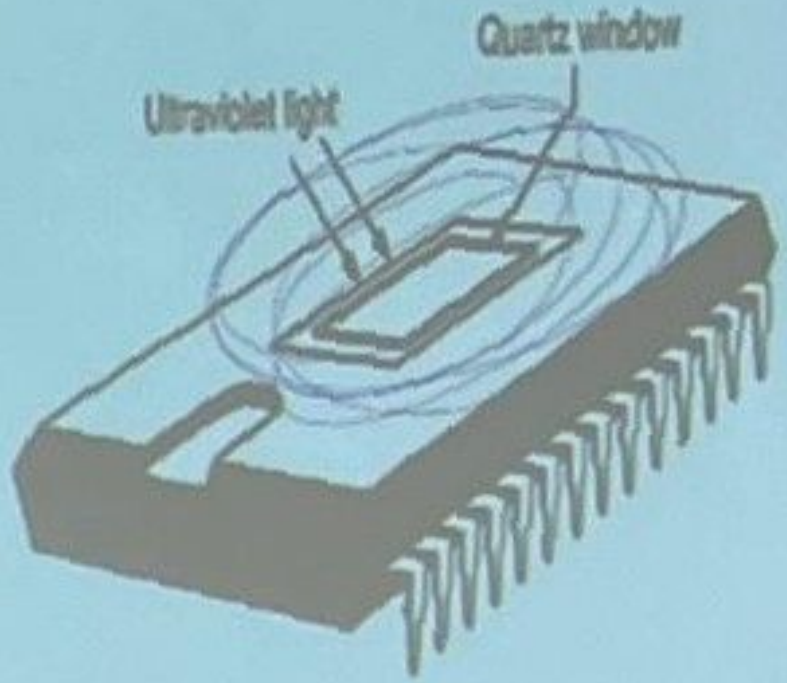
PROM

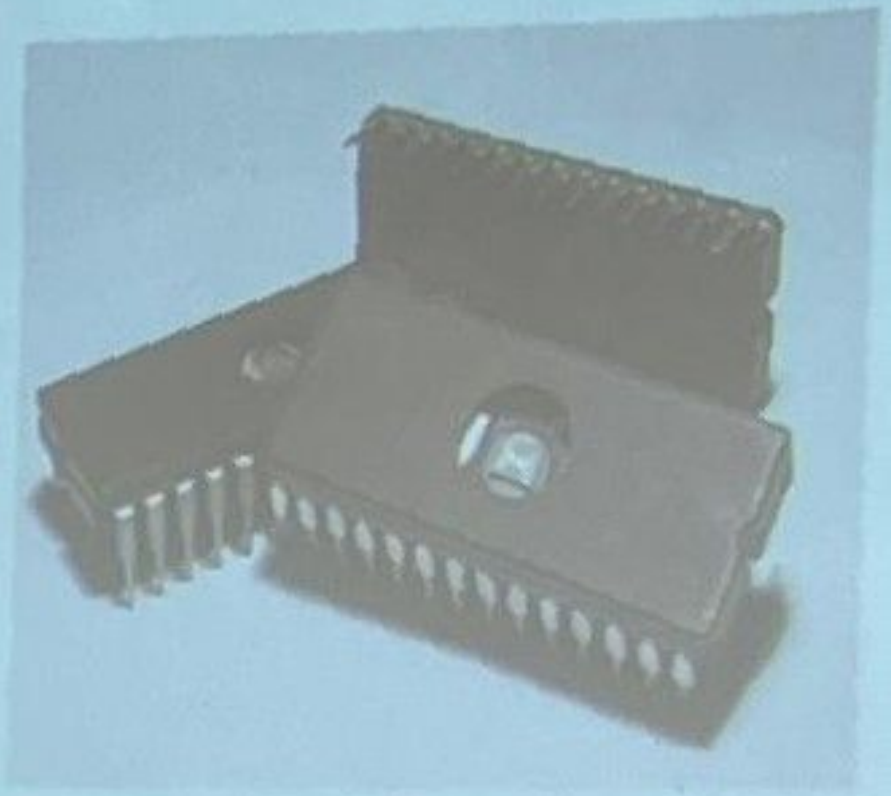
- PROMs are manufactured blank and, depending on the technology, can be programmed at wafer, final test, or in system.
- Blank PROM chips are programmed by plugging them into a device called a PROM programmer.
- For example, you have used Compact disk CD-R in around 2008 this PROM Used in early stages.

EPROM

- *Erasable Programmable Read Only Memory (EPROM) is a form of semiconductor memory can be programmed and then erased at a later time.*
- This is normally achieved by exposing the silicon to ultraviolet light.
- To enable this to happen there is a circular window in the package of the EPROM to enable the light to reach the silicon of the chip.
- EPROMs are generally employed for programs designed for repeated use but that can be upgraded with a later version of a program.

EPROM



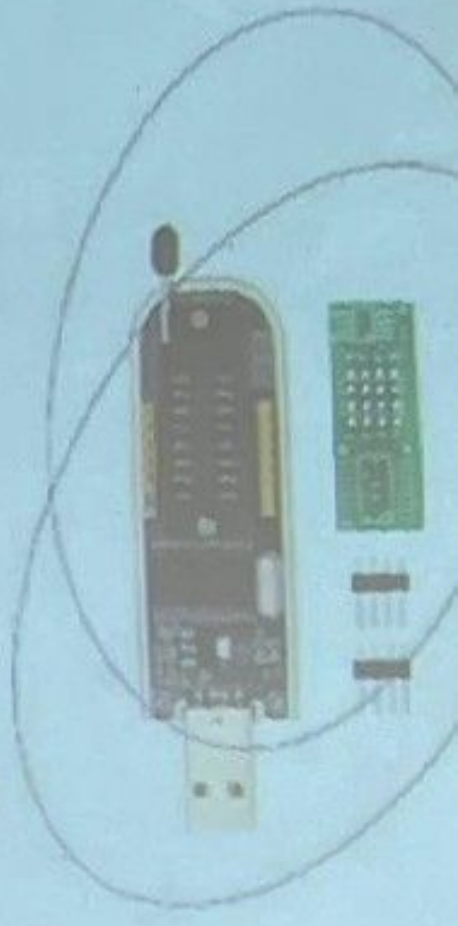


EPROM Uses

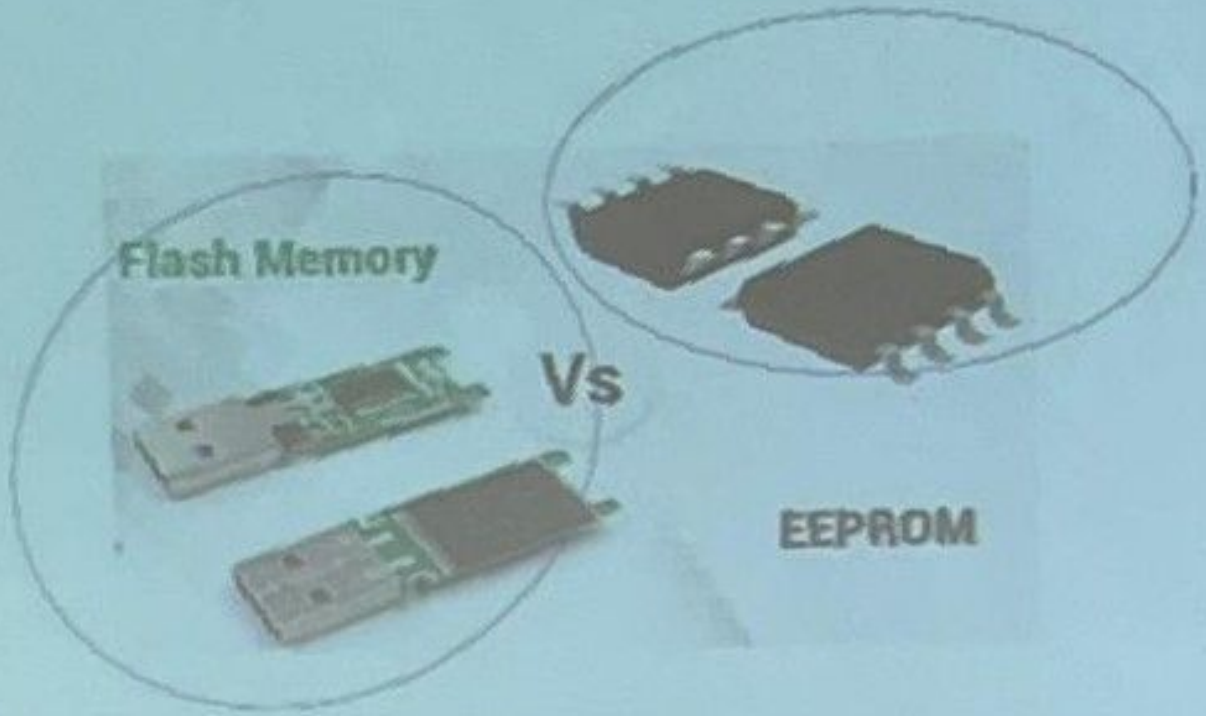
- EPROMs are generally employed for programs designed for repeated use but that can be upgraded with a later version of a program.
- Similar technology is used in CD-RW, DVD-RW.

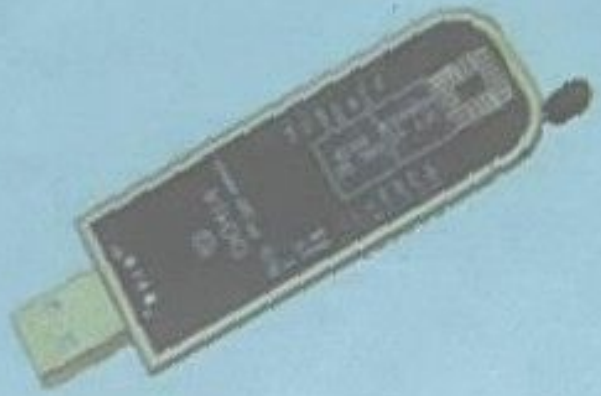
Flash EPROM

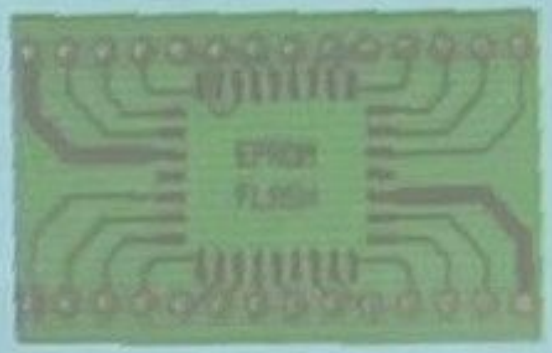
- Flash EPROM or Flash memory is a type of non-volatile memory, specifically a kind of EEPROM (Electrically Erasable Programmable Read-Only Memory), that is erased and programmed in blocks, offering high storage capacity and relatively fast read/write speeds.
- Commonly used in USB drives, SSDs, and memory cards.
- Flash memory erases data in blocks, while EEPROM can erase individual bytes.
- Flash memory generally offers faster read/write speeds than EEPROM
- Flash memory is typically less expensive than EEPROM

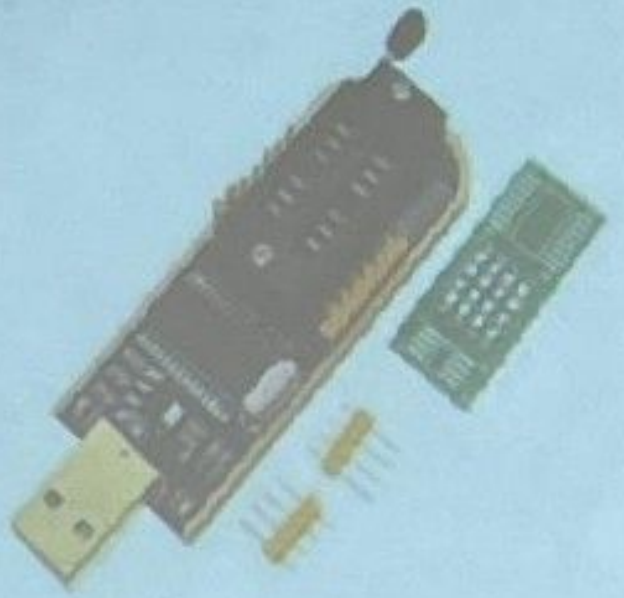


Flash EPROM



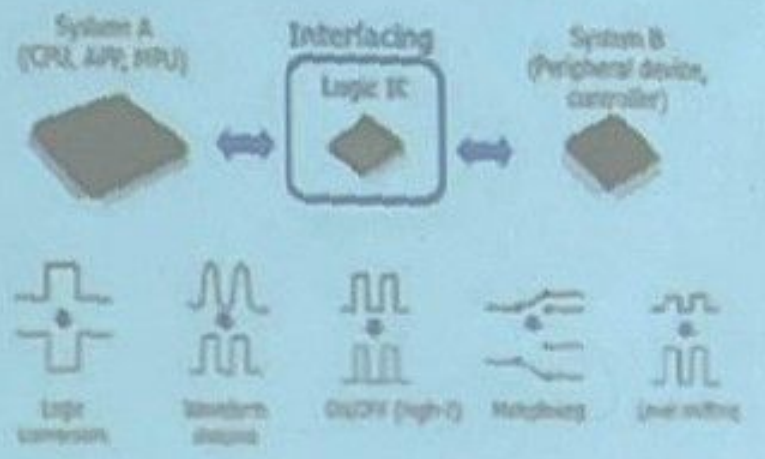




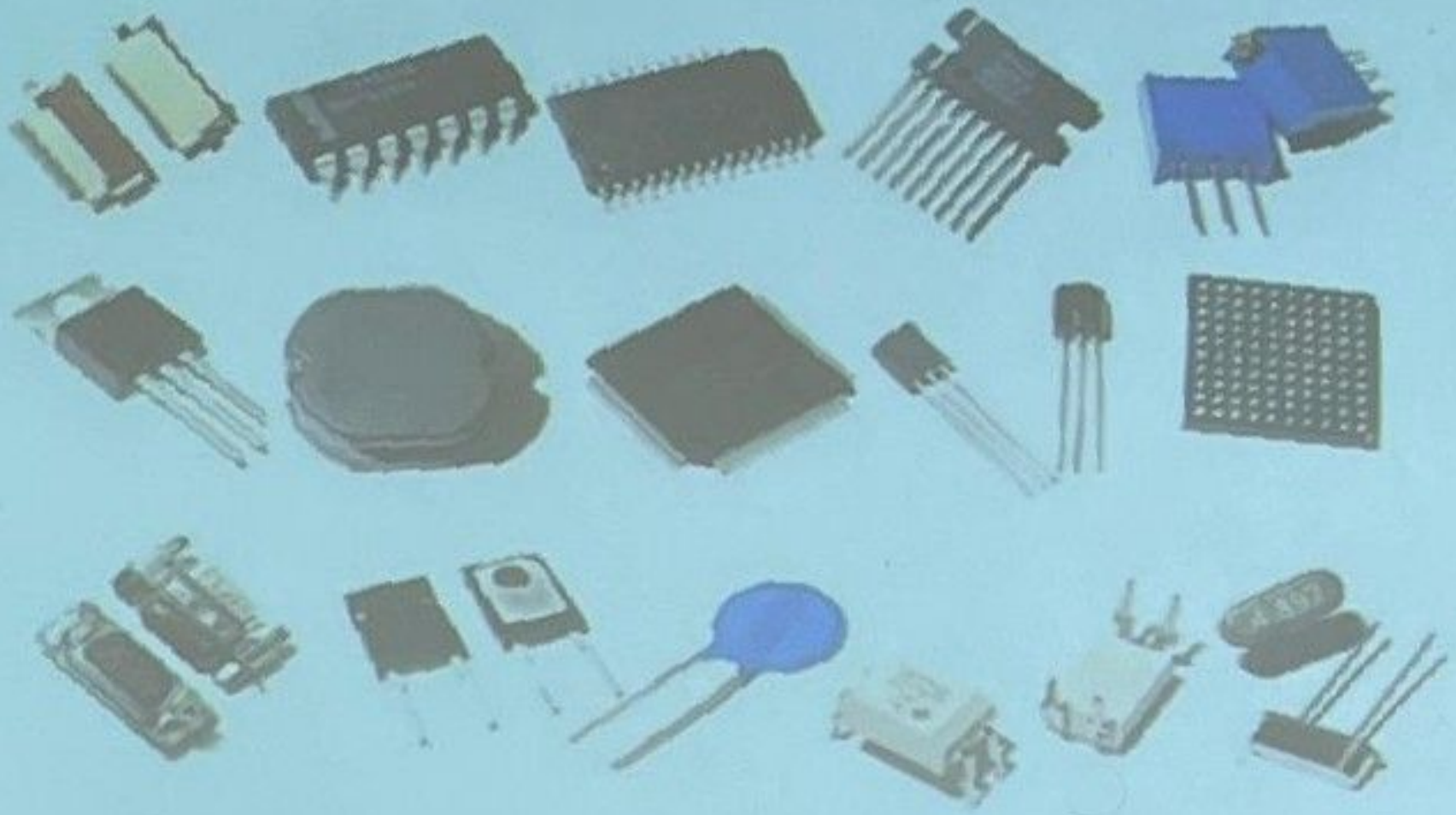


Chip Logic

- As with other integrated circuit products, semiconductor memory comes in packaged chips.
- Each chip contains an array of memory cells.
- The array is organized into W words of B bits each. For example, a 16-Mbit chip could be organized as 1M 16-bit words.
- At the other extreme is the so-called 1-bit-per-chip organization, in which data are read/written 1 bit at a time.



Chip Logic

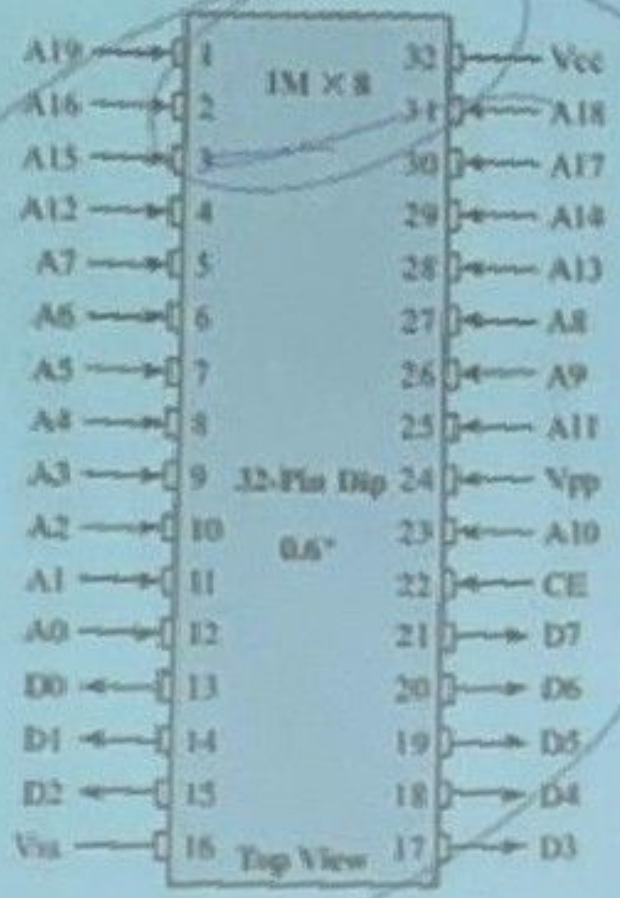


Chip Packaging

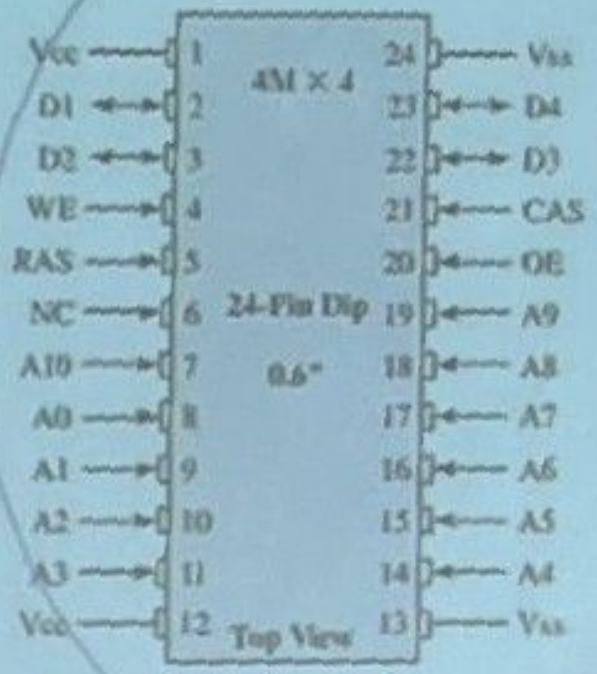
an integrated circuit is mounted on a package that contains pins for connection to the outside world

- An integrated circuit is mounted on a package that contains pins for connection to the outside world.
- Figure 2 shows an example EPROM package.
- The package is an 8-Mbit chip organized as 1Mx8.
- In this case, the organization is treated as a one-word-per-chip package.
- The package includes 32 pins, which is one of the standard chip package sizes.

Chip Packaging Contd...



(a) 8-Mbit EPROM

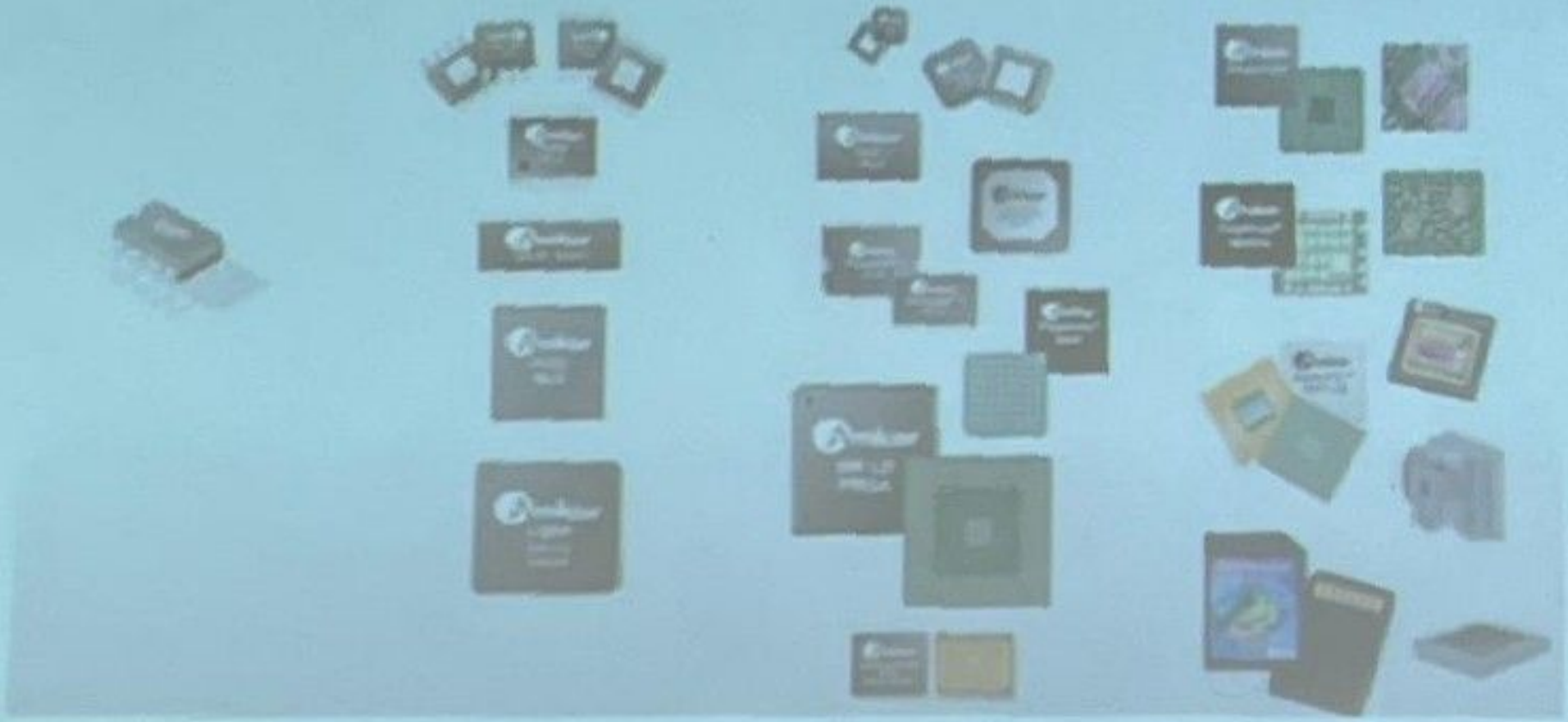


(b) 16-Mbit DRAM

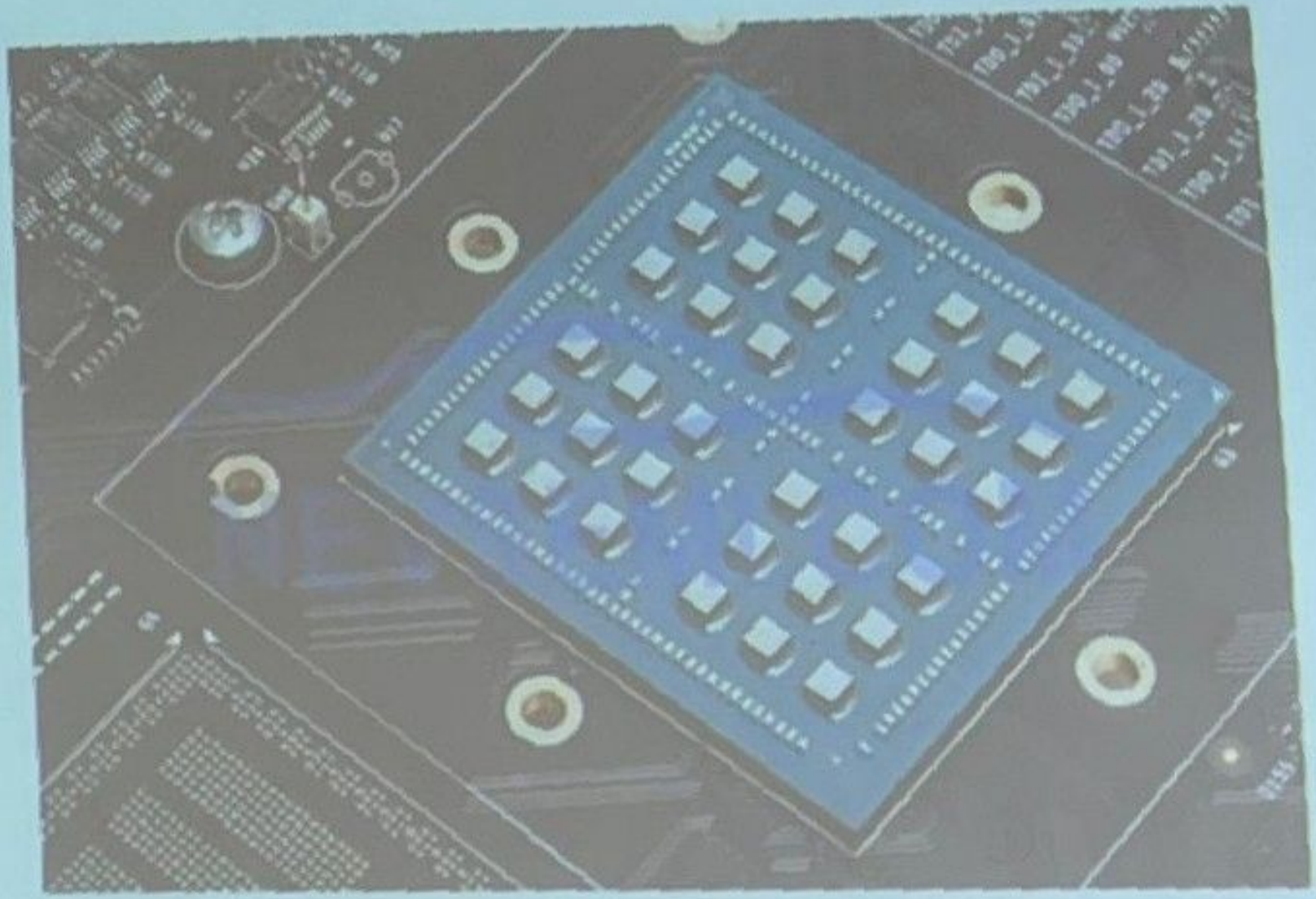
Figure: Typical Memory Package Pins and Signals

Chip Packaging Contd...

1960 – 1985	1986 – 1995	1996 – 2000	2001 – 2005
CDIP, PDIP+	SOIC, PLCC, QFP+	BGA, QFN, SiP+	Modules, Cards, Stack
> 50 pkg types	> 250	> 1000	> 1500



Module Organization

- If a RAM chip contains only 1 bit per word, then clearly we will need at least a number of chips equal to the number of bits per word.
- For 256K words, an 18-bit address is needed and is supplied to the module from some external source (e.g., the address lines of a bus to which the module is attached).
- The address is presented to 8 256Kx1-bit chips, each of which provides the input/output of 1 bit.
- This organization works as long as the size of memory equals the number of bits per chip.

Module Organization